Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-B Congestive Heart Failure

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-02-C01-BD1

Brain dump 1 · foundational retrieval: From a failing left ventricle to alveolar flooding

A competent response contains

· Route: left ventricle with a raised end-diastolic pressure, open mitral valve, left atrium, pulmonary veins, pulmonary capillaries, interstitium, alveoli.

· The pulmonary veins have no valves, so left atrial pressure passes directly back to the pulmonary capillaries; the open mitral valve in diastole is what passes ventricular pressure to the atrium.

· Normal pulmonary capillary pressure is about 7 mmHg against a plasma oncotic pressure of about 28 mmHg, so lymphatics clear the small net filtration easily.

· Pulmonary capillary wedge pressure approximates left atrial pressure and is normally about 6 to 12 mmHg.

· Fluid collects first in the interstitium around vessels and small airways; alveolar flooding follows once capillary pressure exceeds roughly the plasma oncotic pressure and filtration outruns lymph flow.

· A low plasma albumin lowers the oncotic pressure and so lowers the flooding threshold.

· Loss of muscle: ischemic heart disease, the most common cause, through infarction and chronic ischemia.

· Pressure load: systemic hypertension and aortic stenosis, producing concentric hypertrophy and impaired filling.

· Volume load: aortic and mitral regurgitation, which dilate the ventricle.

· Direct muscle damage or stiffening: cardiomyopathies, myocarditis, sustained tachyarrhythmias, alcohol, anthracyclines.

· Mitral stenosis and left atrial myxoma produce every pulmonary sign while the left ventricle stays normal or underfilled, because the obstruction sits at or above the mitral valve.

Distinctions that matter

· A raised wedge pressure reports left atrial pressure, which is not the same as a weak left ventricle: mitral stenosis raises one without the other.

· Interstitial edema and alveolar edema are different stages with different findings, separated by whether filtration has outrun lymphatic clearance.

Common misconceptions

· Saying pulmonary edema begins in the alveoli. Fluid collects in the interstitium first, which is why dyspnea precedes crackles and frothy sputum.

· Treating the flooding threshold as a fixed wedge pressure. It falls with a low albumin and rises when chronic congestion has enlarged the lymphatics.

Where it is taught: Section C01, opening paragraph, causes, and the paragraph on raised left atrial pressure of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C01-BD2

Brain dump 2 · integration and application: Explain three breathing patterns and one blood test

A competent response contains

· Exertional dyspnea: interstitial fluid stiffens the lungs and stimulates juxtacapillary (J) receptors, which drive rapid shallow breathing, and alveolar fluid adds ventilation-perfusion mismatch and hypoxemia.

· Orthopnea: lying flat shifts several hundred millilitres of blood from the legs and splanchnic veins into the thorax, and a ventricle on the flat part of its Frank-Starling curve converts that preload into higher capillary pressure rather than more stroke volume.

· Paroxysmal nocturnal dyspnea: edema fluid from the legs is reabsorbed over one to a few hours while sympathetic support and respiratory centre sensitivity fall during sleep, so the central volume load builds slowly.

· Orthopnea appears within a minute or two because the blood shift is immediate; PND takes hours because it depends on gradual reabsorption of interstitial fluid from the legs.

· Wheeze comes from bronchial wall edema narrowing the airways, which is cardiac asthma.

· Crackles are fine and late inspiratory, from fluid-filled small airways snapping open, and begin at the bases where capillary hydrostatic pressure is highest.

· Forward failure gives prerenal azotemia with a BUN to creatinine ratio above about 20, because enhanced proximal reabsorption and ADH-driven urea reabsorption in the collecting duct accompany the fall in filtration, so urea is retained out of proportion to creatinine.

· Other forward findings: fatigue, cool pale extremities, and confusion in older patients.

· Pleural effusions in this setting are transudates and are often bilateral.

Distinctions that matter

· Orthopnea and PND are both posture related but differ in timing and mechanism: a volume shift against a reabsorption process.

· Creatinine tracks filtration alone, while urea is reabsorbed, which is why the ratio rather than either value identifies the prerenal pattern.

Common misconceptions

· Explaining PND as fluid moving into the chest the moment the patient lies down. That is orthopnea; PND waits for leg edema to be reabsorbed.

· Reading wheeze as evidence of asthma rather than heart failure. Bronchial wall edema wheezes too.

Where it is taught: Section C01, paragraph on dyspnea, orthopnea, and PND, and the final paragraph on crackles and forward findings of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C01-RT01

Retrieval exercise: Rebuild the seven-step congestion sequence

A competent response contains

· Step 1: left ventricular end-diastolic pressure rises.

· Step 2: left atrial and pulmonary venous pressures rise.

· Step 3: capillary filtration exceeds lymphatic clearance, giving interstitial edema.

· Step 4: stiff lungs and J receptor stimulation cause dyspnea on exertion.

· Step 5: lying flat shifts blood into the chest, causing orthopnea.

· Step 6: leg edema is reabsorbed during sleep, causing paroxysmal nocturnal dyspnea.

· Step 7: alveolar flooding gives crackles, hypoxemia, and pulmonary edema.

· At step 3 the capillary hydrostatic pressure now outweighs the plasma oncotic pressure of about 28 mmHg, and the lymphatic route has been exceeded.

· At step 5 several hundred millilitres shift centrally, and a ventricle on the flat part of the Frank-Starling curve gains almost no stroke volume from it, so the pressure appears in the capillaries.

· Long-standing mitral stenosis enlarges the lymphatics so they carry many times normal flow, which moves the flooding step to a much higher pressure, and the chronically raised pressure has also thickened the pulmonary arterioles.

· One day after a large infarct the lymphatics have had no time to adapt, so a wedge pressure that a chronic patient tolerates floods the alveoli.

Distinctions that matter

· The rate of pressure rise matters as much as the level reached, which is why the same wedge pressure means different things acutely and chronically.

· Lymphatic capacity is the variable that decides when interstitial edema becomes alveolar edema.

Common misconceptions

· Writing the sequence as a list of symptoms rather than a sequence of pressures. Each step adds central volume or pressure the ventricle cannot absorb.

· Assuming a chronic patient with a high wedge pressure must be in alveolar edema. Enlarged lymphatics let them tolerate it.

Where it is taught: Section C01, flow diagram on a failing left ventricle to orthopnea and PND of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C02-BD1

Brain dump 1 · foundational retrieval: Right-sided failure from causes to organs

A competent response contains

· Most common cause is left-sided failure: raised pulmonary venous pressure raises pulmonary artery pressure, first passively and then through arteriolar remodeling, and the right ventricle hypertrophies, dilates, and fails, giving biventricular failure.

· Cor pulmonale is right ventricular hypertrophy, dilation, or failure from pulmonary hypertension arising in the lungs or pulmonary vessels, not from left heart or congenital disease.

· Acute cor pulmonale follows massive pulmonary embolism; chronic cor pulmonale follows chronic obstructive pulmonary disease, interstitial lung disease, hypoventilation syndromes, pulmonary arterial hypertension, and chronic thromboembolic disease, through hypoxic pulmonary vasoconstriction and later medial hypertrophy.

· Right ventricular infarction accompanies an inferior infarct from proximal right coronary occlusion and gives hypotension with a raised JVP and clear lungs.

· The right internal jugular vein acts as a manometer for right atrial pressure, normally about 0 to 6 mmHg; with the patient at about 45 degrees, a pulsation more than about 3 to 4 cm above the sternal angle is raised.

· Hepatojugular reflux: abdominal pressure for about 10 to 15 seconds pushes splanchnic blood toward the right heart, and a failing right ventricle cannot accept it, so the JVP stays up while pressure is held.

· Dependent edema needs both raised venous pressure, which increases capillary filtration, and renal sodium and water retention, which supplies the volume.

· Several litres accumulate before swelling is visible, so weight gain comes first; the fluid settles at the ankles in ambulatory patients and the sacrum in bedridden ones.

· It pits because it is a low-protein transudate, unlike lymphedema or the myxedema of hypothyroidism.

· Hepatic acinus: zone 3 is farthest from portal inflow, becomes hypoxic, and atrophies; zone 1 may show fatty change, giving the dark red and tan nutmeg pattern.

· Low output added to congestion gives centrilobular hemorrhagic necrosis; years of congestion give centrilobular fibrosis, called cardiac sclerosis.

· Congestion also produces a tender enlarged liver, ascites, congestive splenomegaly, and bowel wall edema with anorexia.

Distinctions that matter

· Cor pulmonale excludes right ventricular failure caused by left heart disease, which is why the most common cause of right-sided failure is not cor pulmonale.

· Raised venous pressure alone does not make edema; the kidney has to supply the volume, which is why the two requirements are separate.

Common misconceptions

· Calling every raised JVP with ankle edema cor pulmonale. Left-sided failure is the usual cause, and the lungs tell you which.

· Expecting zone 1 to be the hypoxic zone. Zone 1 is closest to portal inflow; zone 3 sits at the end of the gradient and is the one that dies.

Where it is taught: Section C02, causes, the paragraph on neck veins and pitting edema, and the nutmeg liver paragraph of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C02-BD2

Brain dump 2 · integration and application: Three patients with a raised JVP

A competent response contains

· Patient 1: biventricular failure driven by left-sided failure. The left ventricle is failing, pulmonary venous pressure is raised, and the right ventricle has failed against that raised pulmonary artery pressure, so the lungs are congested and the apex is displaced.

· Patient 2: chronic cor pulmonale. The lesion sits in the lungs and pulmonary vessels, so the right ventricle faces a high afterload while left atrial pressure stays normal and the lungs are clear of congestion.

· Patient 3: right ventricular infarction with an inferior infarct from proximal right coronary occlusion. Muscle loss sits in the right ventricle, and the lungs are clear.

· The lung finding follows the position of the lesion in two pumps in series: congestion appears only in the venous bed behind the failing chamber, and only the left heart has the pulmonary veins behind it.

· Patient 2 mechanism: alveolar hypoxia causes hypoxic pulmonary vasoconstriction, and over time the arterioles undergo medial hypertrophy, so the pressure becomes fixed rather than reversible.

· Patient 3 left ventricular filling is reduced for two reasons: right ventricular output falls so less blood crosses the lungs to the left heart, and the dilated right ventricle shifts the septum leftward.

· Volume helps patient 3 because the underfilled left ventricle needs preload and the failing right ventricle must be pushed through the lungs, whereas patient 1 already has a high left atrial pressure and extra volume floods the alveoli.

· Isolated right-sided failure raises systemic venous pressure while pulmonary venous pressure stays normal, which is the full neck veins with clear lungs pattern.

Distinctions that matter

· A raised JVP locates the problem at the right heart but says nothing about the lungs; the lung fields are what separate biventricular failure from isolated right-sided failure.

· Right ventricular infarction and cor pulmonale both give clear lungs, and the difference is lost muscle against a raised afterload.

Common misconceptions

· Assuming any hypotensive patient with an inferior infarct is volume overloaded because the neck veins are full. The left ventricle is underfilled.

· Expecting crackles whenever the JVP is raised. Crackles require raised pulmonary venous pressure, which isolated right-sided failure does not produce.

Where it is taught: Section C02, causes paragraph, the left versus right table, and the stop and predict box of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C03-BD1

Brain dump 1 · foundational retrieval: Two failing ventricles, two sounds, one peptide

A competent response contains

· Ejection fraction is stroke volume divided by end-diastolic volume, normally about 55% to 70%; HFrEF is 40% or lower, mildly reduced is 41% to 49%, and HFpEF is 50% or higher.

· In HFrEF the end-systolic pressure-volume relationship flattens, so the ventricle ejects less and keeps a larger end-systolic volume.

· It adds sarcomeres in series, which is eccentric hypertrophy, and uses the Frank-Starling mechanism at a larger end-diastolic volume.

· Law of Laplace: wall stress is proportional to pressure times radius divided by wall thickness, so a dilated ventricle has high wall stress and high oxygen demand, and the stretched mitral annulus may leak as functional mitral regurgitation.

· Active relaxation needs ATP to return Ca2+ to the sarcoplasmic reticulum through SERCA2a and to release cross-bridges, so ischemia, hypertrophy, and aging slow it.

· Passive stiffness rises with concentric hypertrophy, which is sarcomeres added in parallel, with fibrosis, titin changes, and infiltration such as amyloid, so the end-diastolic pressure-volume relationship steepens and a normal volume needs a high diastolic pressure.

· HFpEF is most common in older adults, women, and patients with hypertension, obesity, diabetes, and atrial fibrillation.

· S3: low-pitched, early diastolic, at the apex, produced when rapid filling from a high-pressure atrium is suddenly halted by a dilated ventricle near its limit of distension; it marks volume overload and high filling pressure, classically in HFrEF, and is normal in children, young adults, athletes, and pregnancy.

· S4: late diastolic, from atrial contraction into a stiff ventricle, so it accompanies HFpEF and disappears in atrial fibrillation.

· BNP is released from stretched ventricular myocytes as proBNP, cleaved into active BNP and inactive NT-proBNP; BNP raises cGMP through the NPR-A receptor and is cleared by neprilysin and NPR-C, while NT-proBNP is cleared mainly by the kidney.

· A BNP below about 100 pg/mL or an NT-proBNP below about 300 pg/mL makes acute heart failure unlikely.

· Levels rise with age, renal failure, atrial fibrillation, and right ventricular strain, and fall with obesity; they are lower in HFpEF because a thick wall lowers wall stress for a given pressure.

Distinctions that matter

· Eccentric and concentric hypertrophy differ in the direction sarcomeres are added, which is why one dilates and the other thickens.

· The S3 reports filling into a distended ventricle and the S4 reports atrial contraction into a stiff one, so the timing in diastole follows the mechanism.

Common misconceptions

· Treating a preserved ejection fraction as a normally functioning ventricle. It empties the fraction it fills, but it fills only at a high pressure.

· Reading any S3 as pathological. It is a normal finding in children, young adults, athletes, and pregnancy.

Where it is taught: Section C03, opening paragraph, HFrEF and HFpEF paragraphs, and the paragraphs on S3 and BNP of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C03-BD2

Brain dump 2 · integration and application: Why the numbers can look reassuring

A competent response contains

· This is heart failure with preserved ejection fraction: her age, long-standing hypertension, obesity, and atrial fibrillation are the classic setting, and hypertension produces concentric hypertrophy with a normal cavity size and a thick wall.

· A normal ejection fraction does not exclude failure because the defect is in filling, not emptying: the steep end-diastolic pressure-volume relationship means a normal volume sits at a high diastolic pressure, which passes back to the pulmonary capillaries.

· BNP is lower than expected for two reasons: a thick wall lowers wall stress for any given pressure, and BNP levels fall with obesity.

· The radiograph shows no cardiomegaly because the cavity is not dilated; a cardiothoracic ratio above 0.5 supports a dilated heart and is often absent in HFpEF, and 0.46 is within normal limits.

· Fast atrial fibrillation takes away the atrial contraction that a stiff ventricle depends on, and it shortens diastole so there is less time to fill, so filling pressure climbs and the lungs flood.

· Radiographic order with wedge pressure: normal at about 6 to 12 mmHg with lower-lobe vessels larger than upper; about 13 to 18 mmHg gives upper-lobe diversion, called cephalization; about 18 to 25 mmHg gives Kerley B lines, peribronchial cuffing, and hazy hila; above about 25 mmHg gives perihilar bat-wing alveolar opacities.

· In failure building over years the lymphatics have enlarged, so the radiographic signs appear at higher pressures, and the signs also lag behind rapid pressure changes.

Distinctions that matter

· Ejection fraction reports emptying and says nothing about the pressure needed to fill, which is the whole of the defect in HFpEF.

· A low BNP argues against acute failure in a thin patient with a dilated ventricle, and argues much less in an obese patient with a thick one.

Common misconceptions

· Using a normal ejection fraction or a normal heart size to rule out heart failure. Neither excludes a stiff ventricle at a high filling pressure.

· Reading a modest BNP as a modest filling pressure. Wall thickness and body habitus both pull the number down.

Where it is taught: Section C03, HFpEF paragraph, BNP paragraph, and the flow diagram on radiographic signs of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C03-CP01

Comparison table: Reduced against preserved ejection fraction

A competent response contains

· Cutoff: HFrEF is an ejection fraction of 40% or lower; HFpEF is 50% or higher, with 41% to 49% called mildly reduced.

· Core defect: HFrEF is reduced contractility; HFpEF is impaired relaxation with increased stiffness.

· Loop: HFrEF has a flatter end-systolic relationship and the loop shifts to the right; HFpEF has a steeper end-diastolic relationship and the loop is narrow at a high diastolic pressure.

· Geometry: HFrEF is dilated with eccentric hypertrophy, sarcomeres added in series, and a normal or thin wall; HFpEF is normal in size with concentric hypertrophy, sarcomeres added in parallel, and a thick wall.

· Causes: HFrEF from myocardial infarction, dilated cardiomyopathy, myocarditis, and chronic volume overload; HFpEF from hypertension, aging, obesity, diabetes, hypertrophic and restrictive cardiomyopathy, and amyloidosis.

· Extra sound: HFrEF gives an S3, from rapid filling halted by a distended ventricle; HFpEF gives an S4, from atrial contraction into a stiff ventricle, when the atria are contracting.

· Dependence on atrial contraction: moderate in HFrEF; high in HFpEF, so atrial fibrillation is poorly tolerated.

· Why filling pressure rises: in HFrEF a large residual volume and continued filling raise end-diastolic pressure; in HFpEF a stiff chamber needs a high pressure even for a normal volume. Both therefore cause the same congestion.

Distinctions that matter

· The two forms differ in whether the ventricle fails to empty or fails to fill, and they do not differ in the congestion they cause.

· An S4 requires organised atrial contraction, which is why it vanishes in atrial fibrillation while an S3 does not.

Common misconceptions

· Putting thick walls with HFrEF and thin walls with HFpEF. Pressure overload thickens and volume overload dilates.

· Assuming HFpEF is the milder condition because the ejection fraction is normal. The filling pressures and the congestion are the same.

Where it is taught: Section C03, HFrEF against HFpEF table of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C04-BD1

Brain dump 1 · foundational retrieval: Lay out the three compensatory systems

A competent response contains

· The baroreceptors and kidney sense arterial filling, not total volume, and cannot tell a failing pump from hemorrhage, so a fall in cardiac output triggers the hemorrhage response.

· Effective arterial blood volume is the portion of blood volume that the arterial baroreceptors and kidney sense as filling the arterial tree; it is low in heart failure even when total volume is expanded, which is why the systems stay switched on.

· Sympathetic: reduced baroreceptor firing raises sympathetic outflow. Beta-1 stimulation raises heart rate and contractility, alpha-1 arteriolar constriction holds arterial pressure, and venoconstriction raises preload.

· Sympathetic harm over months: catecholamines raise oxygen demand and afterload, cause Ca2+ overload, arrhythmias, and myocyte death, and downregulate beta-1 receptors through receptor kinases and beta-arrestin.

· Renin stimuli: low afferent arteriolar pressure, low macula densa NaCl delivery, and renal beta-1 stimulation.

· Angiotensin II at AT1 receptors constricts arterioles, constricts the efferent arteriole more than the afferent, increases proximal Na+ reabsorption, stimulates thirst, stimulates ADH release, and releases aldosterone.

· Aldosterone drives Na+ reabsorption in collecting duct principal cells at the cost of K+ and H+.

· ADH here is driven by baroreceptors rather than osmolality, which is nonosmotic release; it acts at V2 receptors to insert aquaporin-2 and retain free water, so hyponatremia marks advanced disease.

· Retained volume adds little stroke volume because the ventricle sits on the flat part of the Frank-Starling curve, so the extra filling mostly raises venous pressure and causes congestion.

Distinctions that matter

· Total extracellular volume is expanded while effective arterial blood volume is low, and the kidney responds to the second one.

· Osmotic and nonosmotic ADH release have different stimuli; only the nonosmotic route explains retaining water against a low serum sodium.

Common misconceptions

· Describing these systems as a failed or inappropriate response. They are the correct response to the signal they receive, which is arterial underfilling.

· Expecting sodium retention to raise cardiac output. On a flat Frank-Starling curve it mostly raises venous pressure.

Where it is taught: Section C04, opening paragraph and the paragraphs on sympathetic activation, RAAS, and ADH of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C04-BD2

Brain dump 2 · integration and application: Why blocking hormones works and squeezing harder does not

A competent response contains

· Remodeling is the progressive change in ventricular size, shape, and composition after injury: myocytes enlarge and re-express fetal genes, some die by apoptosis, and angiotensin II and aldosterone drive fibrosis.

· In HFrEF the ventricle becomes larger and more spherical, which raises wall stress and pulls the papillary muscles apart, so functional mitral regurgitation appears and the process becomes self-sustaining.

· Blocking beta-1 receptors, angiotensin II, or aldosterone slows remodeling, which shows that the hormones themselves drive progression.

· Raising contractility alone does not slow remodeling, so the ventricle keeps enlarging; the drive to hypertrophy, apoptosis, and fibrosis is hormonal rather than a matter of how hard the muscle squeezes.

· Natriuretic peptides: ANP from the atria and BNP from the ventricles, acting through NPR-A and cGMP.

· Their three opposing actions are natriuresis, vasodilation, and suppression of renin and aldosterone.

· Three reasons they cannot win: receptor downregulation, clearance by neprilysin, and low renal perfusion, which blunts the natriuretic effect.

· Serum sodium is low because nonosmotic ADH release retains water in excess of sodium.

· Urine sodium is low because angiotensin II and aldosterone avidly reabsorb filtered sodium, so the kidney behaves as if the patient were volume depleted even though both sodium and water are in excess.

Distinctions that matter

· Contractility and remodeling are separate targets: one changes the beat, the other changes the chamber.

· A low serum sodium with a low urine sodium is the signature of a kidney reading low effective arterial volume, not of true depletion.

Common misconceptions

· Assuming a drug that makes the heart contract harder must slow the disease. Remodeling is hormone driven, so it continues.

· Expecting a high urine sodium in a patient who is grossly overloaded with salt and water. Aldosterone and angiotensin II hold it back.

Where it is taught: Section C04, paragraph on remodeling and natriuretic peptides, the compensations table, and the stop and predict box of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C04-RT01

Retrieval exercise: Rebuild the neurohormonal cycle

A competent response contains

· Step 1: myocardial injury lowers cardiac output and arterial filling.

· Step 2: baroreceptors and the kidney sense low effective arterial volume.

· Step 3: sympathetic activation, renin release, and nonosmotic ADH release.

· Step 4: short-term support through higher heart rate, contractility, vascular tone, and blood volume.

· Step 5: vasoconstriction raises afterload, and salt and water retention raise preload beyond benefit, causing congestion.

· Step 6: catecholamines, angiotensin II, and aldosterone cause myocyte loss, hypertrophy, and fibrosis.

· Step 7: remodeling lowers pump function further, which lowers output again and returns the loop to step 1.

· The genuinely useful step is step 4; harm begins at step 5 with raised afterload and congestion and continues at step 6 with remodeling.

· Vasoconstriction raises afterload, and salt and water retention raise preload.

· Remodeling hormones: catecholamines cause myocyte death and Ca2+ overload, angiotensin II causes hypertrophy and fibrosis, and aldosterone causes myocardial and vascular fibrosis.

· A response designed for hemorrhage, sustained for months, worsens the pump it is trying to support, because the pump cannot use the extra volume and cannot tolerate the extra resistance.

· An AT1 receptor blocker interrupts the loop at step 3 into 5 and at step 6; a mineralocorticoid receptor blocker interrupts the retention at step 5 and the fibrosis at step 6.

Distinctions that matter

· Preload and afterload are raised by different arms of the same response, so each needs naming separately.

· Short-term support and long-term harm are the same mechanisms at different durations, not different mechanisms.

Common misconceptions

· Drawing the loop as a straight line ending in congestion. It closes, which is why the condition progresses on its own.

· Placing the harm at the sensing step. The sensing is accurate for arterial filling; the harm comes from sustaining the response.

Where it is taught: Section C04, cycle diagram on the neurohormonal cycle of heart failure of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C04-PR01

Prediction exercise: Predict what six months of neurohormonal drive does

A competent response contains

· Systemic vascular resistance rises, so afterload and the work of ejection rise: alpha-1 adrenergic constriction plus angiotensin II at AT1 receptors.

· Preload rises and venous pressure behind both ventricles rises: venoconstriction plus angiotensin II and aldosterone driven sodium retention and ADH driven water retention.

· Beta-1 receptors on the myocyte surface fall in number: sustained catecholamine exposure downregulates them through receptor kinases and beta-arrestin.

· Serum potassium falls because aldosterone reabsorbs Na+ in collecting duct principal cells at the cost of K+ and H+; serum sodium falls because nonosmotic ADH acts at V2 receptors to insert aquaporin-2 and retain water in excess of sodium.

· Interstitial collagen rises: angiotensin II and aldosterone drive myocardial fibrosis.

· The ventricle becomes larger and more spherical, and by the law of Laplace the larger radius raises wall stress, which raises oxygen demand and pulls the papillary muscles apart.

· The ability of natriuretic peptides to offset all of this falls: NPR-A downregulation, neprilysin clearance, and low renal perfusion blunt natriuresis and vasodilation.

Distinctions that matter

· Hypokalemia comes from aldosterone and hyponatremia from ADH, so the two electrolyte changes have separate hormones behind them.

· Receptor downregulation reduces the response to catecholamines without reducing the catecholamine level, which is why the heart is both overstimulated and underresponsive.

Common misconceptions

· Expecting sodium to rise because the body is retaining sodium. Water is retained in greater proportion, so the serum sodium falls.

· Treating natriuretic peptides as a brake that eventually stops the cycle. Their effect is too weak to offset the vasoconstrictor systems.

Where it is taught: Section C04, RAAS and ADH paragraphs, remodeling paragraph, and the compensations table of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C05-BD1

Brain dump 1 · foundational retrieval: Organise the complications by their cause

A competent response contains

· The three headings are high filling pressures, low forward output, and enlarged, scarred, neurohormonally stimulated chambers.

· Acute pulmonary edema sits under high filling pressure: left atrial pressure rises faster than lymphatics can adapt, triggered by ischemia, a hypertensive surge, a fast arrhythmia, acute valve regurgitation, or a sudden salt load, giving severe dyspnea, diffuse crackles, frothy pink sputum, and hypoxemia; the sympathetic surge it provokes raises afterload further.

· The same picture at a normal wedge pressure points to noncardiogenic edema, as in acute respiratory distress syndrome.

· Cardiogenic shock sits under low forward output: systolic pressure below about 90 mmHg for at least 30 minutes, cardiac index below about 2.2 L/min/m2, and wedge pressure above about 15 to 18 mmHg,

classically after infarction of 40% or more of the left ventricle, with a cold wet patient, low mixed venous oxygen saturation, and lactic acidosis.

· Atrial fibrillation sits under chamber change: atrial stretch and angiotensin II and aldosterone driven fibrosis promote it, it removes the atrial kick and shortens diastole, and persistently fast rates can cause tachycardia-induced cardiomyopathy.

· Ventricular tachycardia and fibrillation cause many sudden deaths: scar supports reentry, fewer repolarizing K+ channels prolong the action potential and favor early afterdepolarizations, catecholamines and Ca2+ overload favor delayed afterdepolarizations, and hypokalemia and hypomagnesemia destabilize repolarization.

· Mural thrombi follow Virchow's triad: stasis in a dilated ventricle, akinetic segment, or fibrillating left atrial appendage; endocardial injury over an infarct; and a procoagulant state. They show lines of Zahn and embolize to the brain, gut, kidneys, spleen, and limbs.

· Cardiorenal syndrome has an arterial mechanism, low output and renal vasoconstriction reducing renal flow, and a venous mechanism, raised central venous pressure raising renal venous and interstitial pressure inside the capsule and lowering the gradient for filtration; creatinine and BUN rise and the added sodium retention worsens congestion.

· Cardiac cachexia comes from TNF alpha and interleukin-6 driven proteolysis, gut congestion with anorexia and malabsorption, and higher energy expenditure, and edema can hide the weight loss.

· Profiles: cardiogenic shock is low output, high wedge pressure, high systemic vascular resistance; hypovolemic shock is low output, low wedge pressure, high resistance; early septic shock is high output, low or normal wedge pressure, low resistance.

Distinctions that matter

· Wedge pressure is what separates cardiogenic from hypovolemic shock, since both have a low output and a high resistance.

· Cardiorenal injury is not purely a flow problem; the venous side acts through pressure inside a non-expandable capsule.

Common misconceptions

· Treating every pulmonary edema as cardiac. A normal wedge pressure with the same picture means the capillary leak is the problem.

· Reading a stable weight as evidence against cachexia. Retained fluid masks the loss of muscle.

Where it is taught: Section C05, paragraphs on acute pulmonary edema and cardiogenic shock, arrhythmias, and mural thrombi, cardiorenal syndrome, and cachexia of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C05-BD2

Brain dump 2 · integration and application: Four deteriorations in one patient

A competent response contains

· First occasion: acute cardiogenic pulmonary edema. Left atrial pressure rose faster than the lymphatics could adapt, so alveolar flooding gives frothy pink sputum and diffuse crackles, and the sympathetic surge raises afterload and worsens it.

· Second occasion: noncardiogenic pulmonary edema, as in acute respiratory distress syndrome. A wedge pressure of 10 mmHg is normal, so the fluid cannot be coming from a raised hydrostatic pressure and must come from increased capillary permeability.

· Third occasion: cardiorenal syndrome.

· Third occasion mechanisms: low output with renal vasoconstriction reduces renal blood flow, and the raised central venous pressure raises renal venous and interstitial pressure.

· Because the kidney sits inside a non-expandable capsule, a rise in interstitial pressure opposes glomerular filtration directly, lowering the net filtration gradient even when arterial pressure is adequate; sodium retention then worsens the congestion that caused it.

· Fourth occasion: embolic stroke from a mural thrombus in the dilated, akinetic left ventricle.

· The three conditions are Virchow's triad: stasis in a dilated ventricle or akinetic segment, endocardial injury over the infarcted area, and a procoagulant state.

· On section the thrombus shows lines of Zahn, alternating pale platelet and fibrin layers with darker red cell layers.

· His weight has not risen despite muscle loss because retained fluid replaces the lost tissue mass; the cachexia itself comes from TNF alpha and interleukin-6 driven proteolysis, gut congestion with anorexia and malabsorption, and higher energy expenditure.

Distinctions that matter

· Cardiogenic and noncardiogenic edema look identical at the bedside and are separated by the filling pressure.

· Congestive kidney injury and low-flow kidney injury can occur in the same patient, and the venous mechanism can dominate while the blood pressure is near normal.

Common misconceptions

· Assuming a rising creatinine in heart failure always means the patient needs more volume. A raised venous pressure can be the cause.

· Expecting a mural thrombus to need a hypercoagulable blood state alone. Stasis and endocardial injury are supplied by the failing, infarcted ventricle.

Where it is taught: Section C05, acute pulmonary edema paragraph, and the paragraph on mural thrombi, cardiorenal syndrome, and cachexia of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C05-CA01

Clinical application: Cold, wet, and oliguric three days after an infarction

A competent response contains

· Recognize: cardiogenic shock. He meets all three criteria, with a systolic pressure of 84 mm Hg (below about 90 mmHg), a cardiac index of 1.8 L/min/m2 (below about 2.2), and a wedge pressure of 26 mmHg (above about 15 to 18), after a large anterior infarct.

· Explain, cold mottled skin: low forward output with reflex alpha-1 and angiotensin II driven vasoconstriction shunts blood away from the skin, giving the cold and clammy picture.

· Explain, crackles with a wedge pressure of 26 mmHg: the failing left ventricle has raised left atrial pressure above the plasma oncotic pressure, so filtration outruns lymphatic clearance and the alveoli flood.

· Explain, mixed venous saturation of 44%: with a low output the tissues extract more oxygen from each unit of blood delivered, so the arteriovenous oxygen difference widens and venous blood returns desaturated.

· Explain, raised lactate: oxygen delivery has fallen below tissue demand, so anaerobic metabolism produces lactic acidosis.

· Differentiate: hypovolemic shock gives a low output with a low wedge pressure and a high systemic vascular resistance; early septic shock gives a high output with a low or normal wedge pressure and a low resistance; his profile is low output, high wedge pressure, high resistance.

· Investigate: look for the cardiorenal pattern, with the raised venous pressure lowering the filtration gradient inside the renal capsule alongside low renal flow, a prerenal BUN to creatinine ratio above about 20, and a low urine sodium from angiotensin II and aldosterone; the sodium of 129 mmol/L points to nonosmotic ADH release at V2 receptors retaining water in excess of sodium.

· Decide: the self-reinforcing mechanism is that hypotension lowers coronary perfusion, which worsens ischemia and lowers output further; the organ is the heart itself, whose perfusion depends on the diastolic arterial pressure it generates.

· Justify: coronary flow occurs mainly in diastole and depends on aortic pressure, so a systolic pressure of 84 mm Hg starves the very muscle that must raise it, and cardiogenic shock classically requires loss of 40% or more of the left ventricle, so there is little reserve muscle left. The sympathetic surge raises heart rate and contractility, which raise oxygen demand, and raises afterload, which opposes ejection, so it worsens the ischemic deficit it is responding to.

Distinctions that matter

· A high wedge pressure with a low output is what makes this cardiac rather than hypovolemic, and a high resistance is what makes it cardiac rather than septic.

· Mixed venous saturation reports the balance between delivery and extraction, which is why it falls here and rises in high-output states.

Common misconceptions

· Reading cold clammy skin as a sign that the vessels have failed. The vasoconstriction is intact and intense; the pump is what has failed.

· Treating the hyponatremia as a separate electrolyte problem. It is the expected consequence of nonosmotic ADH release in severe forward failure.

Where it is taught: Section C05, cardiogenic shock paragraph, with ADH and RAAS in C04 and forward failure findings in C01 of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C06-BD1

Brain dump 1 · foundational retrieval: How a high output still leaves a congested patient

A competent response contains

· High-output heart failure is congestive failure in which cardiac output is above normal but still inadequate; the defining feature is low systemic vascular resistance from arteriolar vasodilation, arteriovenous shunting, or high metabolic demand.

· Mean arterial pressure is approximately cardiac output times systemic vascular resistance.

· Sequence: resistance falls, so arterial pressure falls unless output rises; baroreceptors and the kidney sense arterial underfilling exactly as in low-output failure; sympathetic activation, RAAS, and ADH retain sodium and water; blood volume, venous return, and output rise; the persistent volume and rate load eventually raises filling pressures and dilates the ventricles.

· Baseline: 93 mmHg divided by 5 L/min is about 18.6 mmHg per L/min, which is about 18.6 Wood units.

· A large dialysis fistula adds a low-resistance channel in parallel and lowers total resistance to about 10 Wood units.

· To hold the mean arterial pressure near 90 mmHg at 10 Wood units, output must rise to about 9 L/min, since 90 divided by 10 is 9.

· Tissues gain little because much of the extra flow passes through the fistula without reaching capillaries, so venous blood returns well oxygenated and the heart carries the extra work.

· Compressing the fistula raises resistance and arterial pressure, and the baroreflex then slows the heart, which is the Nicoladoni-Branham sign.

· Causes: severe anemia, through hypoxic vasodilation and low blood viscosity; thyrotoxicosis, through high metabolic rate and heat dissipation with T3 relaxing vascular smooth muscle while increasing SERCA2a, alpha myosin heavy chain, and beta-1 receptors and decreasing phospholamban; arteriovenous fistula, a low-resistance shunt bypassing capillaries; wet beriberi, where thiamine deficiency impairs oxidative metabolism so arterioles dilate and the energy-starved myocardium dilates, with Shoshin beriberi the fulminant form with lactic acidosis; and extensive Paget disease, where hypervascular bone behaves like many shunts.

· The three thiamine pyrophosphate dependent enzymes are pyruvate dehydrogenase, alpha-ketoglutarate dehydrogenase, and transketolase.

· Pregnancy, sepsis, cirrhosis, severe obesity, and some myeloproliferative disorders also produce high-output states.

Distinctions that matter

· High output and adequate perfusion are not the same thing: output can be above normal while the tissue or the pressure is still short.

· Lowered resistance and raised demand are two separate routes to the same high-output state.

Common misconceptions

· Assuming a high cardiac output rules out heart failure. The congestion comes from the retained volume, not from the output figure.

· Expecting a normal heart to be immune. It sustains the load for a time, and the volume and rate load eventually raises filling pressures.

Where it is taught: Section C06, opening paragraphs, the worked example on the fistula, and the causes paragraph and table of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C06-BD2

Brain dump 2 · integration and application: Sort two congested patients by their resistance

A competent response contains

· Patient 1 has low-output failure and patient 2 has high-output failure.

· Skin: high resistance from alpha-1 and angiotensin II driven vasoconstriction makes patient 1 cool and pale, while low resistance from dilated arterioles or shunting makes patient 2 warm and flushed.

· Pulse: a low stroke volume into a constricted arterial tree gives a weak pulse with a narrow pulse pressure in patient 1, while a large stroke volume into a low-resistance tree gives a bounding pulse with a wide pulse pressure in patient 2.

· Oxygen: patient 1 has a wide arteriovenous oxygen difference because tissues extract more from each unit of blood delivered, so mixed venous saturation is low; patient 2 has a narrow difference because of shunting or flow in excess of demand, so mixed venous saturation is normal or high.

· Both patients have the same congestive findings, congestion, edema, neurohormonal activation, and an S3, because in both the kidney senses arterial underfilling and retains sodium and water, so venous pressures rise regardless of the output.

· Severe anemia: pallor, a low hemoglobin, and a flow murmur.

· Thyrotoxicosis: weight loss, heat intolerance, tremor, and atrial fibrillation.

· Arteriovenous fistula: a continuous bruit or thrill over the site, with bradycardia on compression.

· Wet beriberi: alcohol use or a poor diet, with edema, neuropathy, and lactic acidosis.

· Paget disease: bone pain, an enlarging skull, warm skin over affected bone, and a raised alkaline phosphatase.

· A heart that was already diseased has less reserve for the extra volume and rate load, so filling pressures rise and the ventricles dilate sooner than in a previously normal heart.

Distinctions that matter

· The resistance, not the output, is what produces the skin, pulse, and oxygen findings, which is why those findings separate the two forms at the bedside.

· Mixed venous saturation and the arteriovenous oxygen difference report the same thing from opposite ends.

Common misconceptions

· Assuming warm flushed hands rule out heart failure. They point to low resistance, which is the high-output form.

· Expecting a high mixed venous saturation to mean the tissues are well supplied. In a shunt the blood never reached the capillaries.

Where it is taught: Section C06, the low-output against high-output table, the causes table, and the stop and predict box of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C07-BD1

Brain dump 1 · foundational retrieval: Rupture to shock, and why the murmur can be quiet

A competent response contains

· The mitral apparatus has anterior and posterior leaflets held by chordae tendineae anchored to the anterolateral and posteromedial papillary muscles, which keep the leaflets closed in systole.

· The anterolateral muscle has a dual supply from the left anterior descending and circumflex arteries; the posteromedial muscle is usually supplied only by the posterior descending artery, which is why it is the vulnerable one.

· Chordal rupture: myxomatous degeneration or mitral valve prolapse, infective endocarditis, blunt chest trauma, and less often rheumatic disease.

· Papillary muscle rupture complicates myocardial infarction about 3 to 7 days afterwards, when necrotic muscle has been removed by inflammatory cells before collagen has formed, so the tissue is at its weakest.

· The posteromedial muscle ruptures with inferior infarcts; rupture of a whole trunk is usually rapidly fatal, while rupture of one head can be survivable.

· Step 1: a chord or papillary muscle ruptures and a leaflet flails into the left atrium.

· Step 2: a large regurgitant volume enters a small, noncompliant left atrium each systole.

· Step 3: left atrial pressure rises steeply, giving a giant v wave on the pressure tracing, because a normal-sized atrium that has never dilated has low compliance and cannot absorb the volume.

· Step 4: pulmonary capillary pressure rises abruptly, giving acute pulmonary edema.

· Step 5: forward stroke volume falls because the undilated ventricle cannot raise total output enough.

· Step 6: hypotension and cardiogenic shock follow, and sympathetic vasoconstriction diverts still more flow backward.

· The murmur may be early systolic, decrescendo, soft, or absent because atrial pressure approaches ventricular pressure in late systole, so the gradient driving the jet disappears before systole ends.

· Heart size on the radiograph is normal because neither chamber has had time to dilate.

· Ventricular septal rupture gives a loud holosystolic murmur at the left lower sternal border and an oxygen saturation step-up from right atrium to right ventricle.

Distinctions that matter

· A loud murmur reports a maintained pressure gradient, so in acute mitral regurgitation a quiet murmur is the dangerous finding rather than a reassuring one.

· Timing of papillary muscle rupture follows the stage of infarct repair, not the size of the infarct.

Common misconceptions

· Expecting a loud holosystolic murmur in acute mitral regurgitation. That belongs to the chronic lesion and to septal rupture.

· Reading a normal heart size on the radiograph as evidence against a severe valve lesion. Sudden lesions give no time for dilation.

Where it is taught: Section C07, opening paragraph, causes, hemodynamics and findings, and the flow diagram on rupture to shock of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C07-BD2

Brain dump 2 · integration and application: Work the volumes behind a normal ejection fraction

A competent response contains

· Total stroke volume is 120 minus 30, which is 90 mL.

· Ejection fraction is 90 divided by 120, which is 75%.

· Forward stroke volume is 90 minus 50, which is 40 mL.

· Regurgitant fraction is 50 divided by 90, which is about 56%; it is the backward share of total stroke volume.

· Forward cardiac output is 40 mL times 100 per minute, which is 4.0 L/min, below the 5.0 L/min he had before rupture at a heart rate of 72 per minute.

· The ejection fraction rose because the ventricle now empties into the low-pressure left atrium as well as the aorta, so end-systolic volume fell from 50 mL to 30 mL.

· Ejection fraction measures the fraction of the ventricle's contents ejected anywhere, so it overstates forward performance whenever part of the stroke volume goes backward.

· Those 50 mL per beat enter a small, noncompliant left atrium, so atrial pressure rises steeply with giant v waves and the wedge pressure climbs sharply, flooding the alveoli.

· A patient with fifteen years of mitral regurgitation has a dilated, highly compliant left atrium and a dilated left ventricle with eccentric hypertrophy, which this patient lacks.

· A compliant atrium absorbs the same regurgitant volume with only a modest pressure rise, so there is gradual exertional dyspnea rather than sudden pulmonary edema, and the dilated ventricle maintains forward output for years by generating a larger total stroke volume.

Distinctions that matter

· Total and forward stroke volume are different quantities, and ejection fraction is calculated from the first one.

· Compliance, not regurgitant volume alone, decides how high the atrial pressure rises, which is why the acute and chronic lesions present so differently.

Common misconceptions

· Taking a high ejection fraction as proof of good pump performance. In mitral regurgitation it can be high while forward output is failing.

· Assuming chronic mitral regurgitation is simply a milder leak. The adaptations of the atrium and ventricle are what make it tolerable.

Where it is taught: Section C07, hemodynamics paragraph, the worked example on ejection fraction, and the acute against chronic table of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C02-Q01

Board-style question: Hypotension with a raised JVP and clear lungs

Correct answer: B. Reduced left ventricular filling because right ventricular output has fallen and the septum has shifted leftward

Why it is right: Right ventricular infarction lowers right ventricular output so less blood crosses the lungs to the left heart, and the dilated right ventricle shifts the septum leftward, reducing left ventricular filling and therefore stroke volume; systemic venous pressure rises while pulmonary venous pressure does not, which gives the raised JVP with clear lungs.

A is wrong: Alveolar flooding from a rapid rise in pulmonary capillary pressure would give crackles and radiographic opacities, and the lungs are clear by both.

C is wrong: A fall in systemic vascular resistance would give warm flushed extremities with a wide pulse pressure, which belongs to high-output or septic states rather than an occluded right coronary artery.

D is wrong: Acute regurgitation into a small noncompliant left atrium raises pulmonary capillary pressure and causes sudden pulmonary edema, so the lungs would not be clear.

A competent response contains

· ST elevation in II, III, and aVF marks an inferior infarct, usually from proximal right coronary occlusion, which also supplies the right ventricle.

· A raised JVP with a sustained response to abdominal pressure and clear lungs on auscultation and radiograph places the problem in the right heart with normal pulmonary venous pressure.

Distinctions that matter

· Hypotension with a raised JVP and clear lungs separates right ventricular infarction from left-sided or biventricular failure.

Common misconceptions

· Reading full neck veins as volume overload. The left ventricle here is underfilled.

Where it is taught: Section C02 of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C06-Q02

Board-style question: Congestion with warm extremities on dialysis

Correct answer: A. A large share of the cardiac output returns to the heart without passing through capillaries

Why it is right: A large fraction of the cardiac output passes through the fistula from artery to vein without entering capillaries, so it unloads no oxygen and returns to the heart still saturated, which narrows the arteriovenous oxygen difference and raises mixed venous saturation.

B is wrong: Greater tissue extraction widens the arteriovenous oxygen difference and lowers mixed venous saturation, which is the low-output pattern and the opposite of her figure.

C is wrong: Pulmonary edema limiting alveolar oxygen uptake would lower arterial saturation and, with unchanged extraction, lower rather than raise the mixed venous value.

D is wrong: A reduced hemoglobin concentration lowers oxygen carrying capacity and drives greater fractional extraction, so mixed venous saturation would fall; her finding also has a direct anatomical explanation at the fistula.

A competent response contains

· A large arteriovenous fistula is a low-resistance channel in parallel with the systemic circulation, so systemic vascular resistance falls and cardiac output rises to hold the arterial pressure.

· Warm hands, a bounding pulse with a wide pulse pressure, an output of 9 L/min, and congestion with an S3 are the high-output pattern.

Distinctions that matter

· Mixed venous saturation is high when flow bypasses capillaries and low when tissues extract more from a small delivery.

Common misconceptions

· Assuming a high mixed venous saturation means the tissues are well supplied. Shunted blood never unloaded oxygen.

Where it is taught: Section C06 of the CV-18-B Congestive Heart Failure notes.